SCMS SCHOOL OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MedLedger : A Secure Blockchain-Enabled Federated Learning Framework for Smart Electronic Health Records

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INTRODUCTION

- Electronic Health Records (EHRs) require secure, private, and efficient data sharing across healthcare institutions.
- Centralized healthcare systems are vulnerable to data breaches, single points of failure, and limited interoperability.
- **MedLedger** integrates Blockchain, Federated Learning, and Edge Analytics to provide secure and privacy-preserving EHR management.
- The platform enables decentralized patient record management, collaborative AI model training, and blockchain-based auditability without sharing sensitive patient data.
- It enhances patient privacy, prevents cyber and model poisoning attacks, improves collaboration between hospitals, and supports intelligent healthcare decision-making.

OBJECTIVE OF THE PROJECT

- **Develop a secure and privacy-preserving Electronic Health Record (EHR) management platform** using Blockchain and Federated Learning to protect sensitive patient data.
- **Enable decentralized and intelligent collaboration between healthcare institutions** without sharing raw patient records, ensuring data privacy while improving AI-driven healthcare insights.
- **Enhance trust, security, and transparency** through blockchain-based auditability, secure model verification, and protection against data breaches and model poisoning attacks.

MOTIVATION OF THE PROJECT

- **Growing concerns over patient data privacy** due to increasing cyberattacks, data breaches, and unauthorized access to Electronic Health Records (EHRs).
- **Limitations of centralized healthcare systems**, including single points of failure, poor interoperability, and inefficient data sharing between healthcare institutions.
- **Need for secure collaborative AI in healthcare**, where hospitals can improve disease prediction models without exposing sensitive patient information.
- **Lack of transparency and trust** in healthcare data management, making it difficult to verify the integrity of medical records and AI model updates.
- **Demand for intelligent, privacy-preserving healthcare systems** that combine Blockchain, Federated Learning, and Edge Analytics to enable secure, transparent, and efficient patient care.

CHALLENGES AND CONSIDERATIONS

- **Ensuring patient data privacy and security** while enabling collaborative AI model training across multiple healthcare institutions.
- **Managing blockchain scalability, transaction latency, and computational overhead** to support efficient real-time healthcare applications.
- **Protecting the system against cyber threats, data leakage, and model poisoning attacks** while maintaining the integrity of Electronic Health Records.
- **Integrating the platform with existing Hospital Information Systems (HIS) and Electronic Health Record (EHR) standards** to ensure interoperability.
- **Balancing privacy and AI performance** by optimizing Federated Learning, Differential Privacy, and Blockchain mechanisms.
- **Providing a user-friendly and compliant platform** that meets healthcare regulations (such as HIPAA/GDPR) while remaining easy for doctors, administrators, and patients to use.

ROAD MAP OF THE PROJECT (GANTT CHART)

MODULE	ACTIVITIES COVERED	DURATION
RESEARCH AND DESIGN	Literature Survey,Requirement Analysis, System Design	Jul - Oct
BASELINE EVALUATION	Evaluate existing EHR and Blockchain systems.	Sep - Nov
Blockchain & Federated Learning Implementation	Develop and integrate Blockchain, Federated Learning, and privacy-preserving mechanisms for secure Electronic Health Record (EHR) management..	Nov-Jan
TESTING AND APPLICATION	Integrate the web application, Blockchain, Federated Learning, and database; perform functional, security, and performance testing.	Jan - Apr
DOCUMENTATION AND DISSEMINATION	Research paper writing,Documentation, Final Demonstration	Nov-mar

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THANK YOU